

Production Order List — CSV Export — Screen Processing Design (画面処理設計)

002_DD-SPD-CSV — elaborates [002_DD-CSV](#), implements [002_BD-CSV](#) E-30–E-33 and FN-043, requirements REQ-085 and REQ-089.

Document control (改版履歴)

Field	Value
Document ID	002_DD-SPD-CSV
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-016
Created by	Claude (for ThongTM)
Created date	2026-10-07
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Version	Date	Author	Revision content
1	2026-10-07	Claude (for ThongTM)	Initial creation

Overview and process list

Field	Value
Screen / file name	<code>ExportCsvButton.tsx</code> , <code>ProductionOrderListPage.tsx</code> , <code>api.ts</code> , <code>lib/apiClient.ts</code> , <code>lib/download.ts</code> ; backend <code>ProductionOrdersController.Export</code>
Overview	What happens, step by step, from a press of 「CSV出力」 (Export CSV) to the saved file or the shown error, and how the controls reset when the list view changes

Process IDs continue 002_DD-SPD (P-10–P-15).

No	Process name	Overview	Notes
1	<code>ExportCsvButton</code> — P-16 Export from the list	Activation, limit pre-check, busy state, request, save, announce or map the error	E-30, E-31, E-32
2	<code>ProductionOrderListPage</code> — P-17 Reset on view change	Re-mount the export controls whenever the applied view changes	E-33
3	<code>exportOrdersCsv</code> / <code>apiClient.getFile</code> — P-18 File request	Build the URL, fetch, keep the session/error conventions, read blob, name and count	API-PO-05
4	<code>saveFile</code> — P-19 Save	Hand the blob to the browser as a download	
5	<code>ProductionOrdersController.Export</code> — request boundary	Validate, call the service, return the CSV result or Problem Details	Server side in 002_DD-FN-CSV

Reference documents

No	Document	Purpose / use	Notes
1	002_BD-CSV	Events E-30–E-33, items 28–30, messages	Version 2
2	002_DD-CSV	Modules 1–6, states, screen items	Version 1
3	002_DD-API-CSV	Request, headers (<code>Content-Disposition</code> , <code>X-Total-Count</code>), status codes	Version 1
4	002_DD-FN-CSV	Server processing behind P-18	Version 1
5	002_DD-SPD	P-10–P-15, the list's load/query and view-state processing that P-17 hooks into	Unedited

Processing design

1. `ExportCsvButton` — P-16 Export from the list

Field	Value
Detail	Handles activation of item 28 and drives items 29 and 30 through the states of 002_DD-CSV
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Processing overview: local state `phase: 'ready' | 'exporting'` and `notice: { kind: 'success', count } | { kind: 'error', ids: string[] } | null`, plus a `busyRef` set synchronously so a fast double activation cannot start two requests. An `AbortController` lives for the component's lifetime; unmounting aborts any request in flight (P-17).

Focus while busy: the button is **not** given the HTML `disabled` attribute during an export, because a focused element that becomes `disabled` can drop keyboard focus to the page body, which would break "focus stays on the button" (002_BD-CSV E-30). It gets `aria-disabled="true"` and `aria-busy="true"`, the disabled styling, and the click handler returns immediately while busy. This refines 002_DD-CSV's screen item definition, which says `disabled`; see "Unresolved decisions". The `disabled` prop of module 1 (list query in flight) still uses the real attribute, because then the button is not the focused control.

Used components / services

No	Name	Overview	Notes
1	<code>exportOrdersCsv</code> (P-18)	Request and response reading	
2	<code>saveFile</code> (P-19)	Download	
3	<code>exportHint</code> , <code>message</code> , <code>labels.list</code>	Texts	002_DD-CSV catalog
4	<code>ApiError</code>	Error classification	<code>lib/apiClient.ts</code>

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Render: button 「CSV出力」 with <code>aria-describedby="export-hint"</code> , hint (item 30), message container (item 29, <code>role="status"</code> , empty)	<code>disabled</code> prop true → real <code>disabled</code>	<code>exportHint(view, total)</code>	Ready
2	Activation (click, or Enter/Space on the button)	<code>busyRef.current</code> → return (no second request)	—	—
3	Row-limit pre-check (V-14)	<code>total > 10_000</code> → <code>notice = error [MSG-E024]</code> ; stop. No request	—	Failed (limit); alert rendered
4	Enter busy: <code>busyRef = true</code> , <code>phase = 'exporting'</code> , <code>notice = null</code>	—	—	Button shows 「出力中…」, <code>aria-disabled</code> , <code>aria-busy</code> ; focus unchanged
5	Request	—	<code>exportOrdersCsv(view, controller.signal)</code> (P-18)	—
6	Success: save the file, then announce	—	<code>saveFile(blob, fileName)</code> (P-19)	<code>notice = success(count)</code> → <code>role="status"</code> text MSG-I009 with <code>formatNumber(count)</code>

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
7	Failure: classify	<code>AbortError</code> (unmounted) → do nothing (component is gone). <code>ApiError</code> 401 → nothing here; <code>ApiClient</code> already cleared the session and <code>ProtectedRoute</code> redirects. 403 → [MSG-E020]. 422 with <code>problem.code === 'MSG-E024'</code> → [MSG-E024]. 400 with <code>problem.errors</code> → the message IDs in <code>errors</code> , flattened and de-duplicated (as the list page does). Anything else (other status, network <code>TypeError</code> , body read failure) → [MSG-E025]	—	Failed; <code>role="alert"</code> paragraph with <code>ErrorIcon</code> and the joined messages
8	Leave busy (<code>finally</code> , unless unmounted)	—	—	<code>busyRef = false</code> , <code>phase = 'ready'</code> ; button 「CSV出力」 again; focus unchanged

Notes:

- The success text uses the server's `X-Total-Count`, not the list's `total`, so a change between the list query and the export is reported truthfully (002_BD-CSV "Data changed during export").
- Errors are rendered as a separate `role="alert"` element inside item 29 so they are announced assertively; the success text replaces the content of the always-mounted `role="status"` container so it is announced politely.

2. `ProductionOrderListPage` — P-17 Reset on view change

Field	Value
Detail	Places module 1 and guarantees an old export state never describes a new view
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Processing overview: the page already derives `viewKey = JSON.stringify(view)` for its query effect (002_DD-SPD P-10). It renders `<ExportCsvButton key={viewKey} view={view} total={page.total} disabled={loading} />` inside the `page && page.total > 0` branch. A new applied view (Search, Clear, sort, page, page size, Back/Forward — P-11 to P-14) changes the key, and a running list query removes the branch (loading), so the component unmounts: its `AbortController` aborts any export in flight, and item 29 disappears. When the query finishes, a fresh instance renders with the new hint. Page and page-size changes also re-mount it; harmless, and it keeps one rule for all view changes.

Used components / services: `ExportCsvButton` (P-16).

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	List query starts (P-10)	—	—	Branch not rendered → export controls unmounted; in-flight export aborted
2	List query succeeds	<code>total ≥ 1</code> → render module 1 with the new key; <code>total = 0</code> → empty or no-match panel, no module 1	—	Ready with the new hint, or Hidden
3	List query fails	—	—	Error banner (P-15); no module 1

3. `exportOrdersCsv` / `apiClient.getFile` — P-18 File request

Field	Value
Detail	Fetch API-PO-05 with the list's session and error conventions and return the file
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Processing overview: `apiClient` gains `getFile(url, signal) → Promise<Response>`: the same `fetch` as `request (credentials: 'same-origin', signal)` with `Accept: 'text/csv, application/problem+json'`; on a non-2xx response it does exactly what `request` does (401 → unauthorized handler; throw `ApiError(status, readProblem(response))`), and on 2xx it returns the `Response` unread. This keeps one place that knows about 401 handling and Problem Details (002_DD-CSV X-2). `exportOrdersCsv` in `api.ts` builds the query with `toSearchParams(view)`, deletes `page` and `pageSize`, and reads the response.

Used components / services

No	Name	Overview	Notes
1	<code>toSearchParams</code>	Same serializer as the URL and the list query	Unchanged
2	<code>apiClient.getFile</code>	New export from <code>lib/apiClient.ts</code>	Shares <code>readProblem</code> and the unauthorized handler

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	<code>params = toSearchParams(view); params.delete('page'); params.delete('pageSize')</code>	—	<code>toSearchParams</code>	Only filters and non-default sort remain
2	URL <code>/api/production-orders/export</code> plus <code>?{params}</code> when not empty	—	—	—

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
3	<code>response = await getFile(url, signal)</code>	non-2xx → <code>ApiError</code> thrown (P-16 step 7)	API-PO-05	—
4	<code>blob = await response.blob()</code>	Connection aborted by the server mid-stream (002_DD-FN-CSV §4) → <code>TypeError</code> → P-16 maps to MSG-E025	—	Whole file in memory (≤ 10,000 rows, a few MB)
5	<code>count = Number(response.headers.get('X-Total-Count'))</code>	Missing or not a non-negative integer → treat as protocol error → MSG-E025 (never announce a wrong count)	—	—
6	<code>fileName</code> from <code>Content-Disposition: filename*=UTF-8'...'</code> decoded with <code>decodeURIComponent</code> ; else quoted <code>filename="..."</code> ; else fallback 製造指示一覧_{YYYYMMDD-HHmm}.csv built from the browser clock	Any <code>/</code> , <code>\</code> or control character removed from the name	—	—
7	Return <code>{ blob, fileName, count }</code>	—	—	—

4. `saveFile` — P-19 Save

Field	Value
Detail	Trigger the browser download of a blob under a name
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Processing overview: no library; works in Chromium, Firefox and Safari, PC and SP.

Used components / services: none.

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	<code>url = URL.createObjectURL(blob)</code>	—	—	—
2	Create <code></code> (not attached to the visible layout, <code>tabIndex = -1</code>), <code>click()</code> it	—	—	Browser download starts (Playwright <code>download</code> event)
3	<code>setTimeout(() => URL.revokeObjectURL(url), 0)</code>	—	—	Memory released after the browser took the file

5. ProductionOrdersController.Export — request boundary

Field	Value
Detail	[HttpGet("export")] on the existing controller; class-level policy and NoStore apply
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Processing overview: the HTTP edge only; all processing is in 002_DD-FN-CSV.

Used components / services: ProductionOrderListQuery.TryCreate , ProductionOrderService.ExportAsync , CsvExportResult , ProductionOrderProblems .

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Bind [FromQuery] ProductionOrderExportRequest (strings only, like the list)	—	—	—
2	Map to ProductionOrderListRequest with Page = null , PageSize = null ; validate	Invalid → 400 Problem Details	TryCreate	400
3	ExportAsync(query, userId, ct)	Invalid → 400; RuleViolation → 422 MSG-E024	002_DD-FN-CSV \$1	400 / 422
4	Ok(export) → return new CsvExportResult(export, ...)	—	002_DD-FN-CSV \$4	200, streamed CSV

The route export cannot collide with {id:guid} , because export is not a GUID.

Unresolved decisions

- 1. **Busy button uses aria-disabled , not disabled (proposed, WI-016 DEC-017).** 002_DD-CSV version 1 (approved) specifies disabled + aria-busy while exporting. To keep keyboard focus on the button as 002_BD-CSV E-30 requires, this document uses aria-disabled="true" + aria-busy="true" with a click guard and the same disabled styling. If accepted, 002_DD-CSV gets a version 2 with this one-line change in its screen item definition and state table, and its PDFs are re-rendered; no other document changes.
- 2. Everything else: none. apiClient.getFile (answering 002_DD-CSV's open question about raw responses), the X-Total-Count validation and the file-name sanitising are decided here.